

PROJECT REPORT

Inferring Class - Differentiating Patterns to Transform AI Voices into Human - Like Speech

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Under the Supervision of

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Certificate

It is certified that the project report titled **Inferring Class - Differentiating Patterns to Transform AI Voices into Human - Like Speech** is a report of the project work done by the team of students comprising **Ms. Aishwarya Dhanda, Ms. Shreya Jaisal, Mr. Alakshendra Vashisht, Mr. Pranjal Gupta** and **Mr. Jaydeep Gupta** under the undersigned supervisor for the fulfilment of requirement of project in **Semester VI** of the B.Tech. programme, Faculty of Technology, University of Delhi, India, and that all the members of the team have sufficiently contributed to the project.

To the best of the knowledge and belief of the supervisor, this project

1. embodies the work of candidate(s) herself / himself,
2. fulfills the requirements of **Project** for Semester VI

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Declaration

We hereby declare that the project work presented in this report, titled **Inferring Class - Differentiating Patterns to Transform AI Voices into Human - Like Speech**, is an original piece of work carried out by us. This work has not been submitted to any other university or institute for the award of any degree, diploma, or certification.

We further affirm that all sources of information used in this report have been duly acknowledged, and the work presented reflects our genuine effort toward the successful completion of the requirements for the **Internship, Apprenticeship, Project and Community Outreach (IAPC)**.

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Abstract

The rapid advancement of generative artificial intelligence has significantly enhanced the realism of synthetic speech produced by modern **Text-to-Speech (TTS)** and **Voice Conversion (VC)** systems. While these systems can closely mimic human pronunciation and fluency, fundamental differences persist due to the contrast between **biological speech production** and **AI-driven synthesis**. Human speech is inherently characterized by **natural variability**, including **jitter**, **shimmer** and **articulatory dynamics**, whereas AI-generated speech relies on **mathematically modeled approximations** that often lack these micro - level irregularities. This increasing realism poses critical security challenges, including audio deepfakes, biometric spoofing, and misinformation, thereby necessitating robust and interpretable detection frameworks.

This study presents a **dual - pipeline machine learning framework** designed to both classify and analyze AI - generated versus human speech. A **custom dataset comprising approximately 2000 audio samples**, evenly distributed between synthetic and real speech across **English** and **Hindi languages**, was curated. A standardized **preprocessing pipeline** was applied, including resampling, normalization, mono conversion, and fixed-duration segmentation.

The first pipeline utilizes **Mel spectrogram** representations as input to **Convolutional Neural Networks (CNNs)**, with separate models trained for each language to eliminate linguistic bias. These models not only perform classification but also generate **high - dimensional embeddings** that capture **temporal** and **spectral speech** characteristics. The second pipeline adopts a **hybrid feature extraction approach**, combining **self - supervised representations** derived from **Wav2Vec2** with **handcrafted digital signal processing (DSP) features**. This approach explicitly targets low - level acoustic artifacts such as phase inconsistencies, spectral irregularities, and the absence of biological variations commonly found in synthetic speech.

To enhance interpretability, both pipelines incorporate **feature - space analysis techniques**. Dimensionality reduction using **t - SNE (t - Distributed Stochastic Neighbor Embedding)** is employed to visualize clustering patterns, while **cosine similarity** measures quantify the separation between AI-generated and human speech representations. By integrating deep learning - based feature learning with signal - level analysis, the proposed framework ensures that classification decisions are grounded in meaningful acoustic distinctions rather than superficial patterns.

1. Introduction

1.1 Background and Context

Speech has long been a primary mode of human communication, and its digital processing has evolved significantly with advances in machine learning and artificial intelligence. Early speech synthesis systems were limited in their ability to produce natural - sounding audio, often generating robotic and monotonous outputs. However, recent developments in deep learning have led to the emergence of highly sophisticated TTS and VC systems capable of generating speech that closely resembles human voices in terms of fluency, clarity, and expressiveness. These systems typically rely on neural architectures that convert textual or phonetic inputs into intermediate acoustic representations, such as Mel spectrograms, which are then transformed into audio waveforms using neural vocoders.

At the same time, human speech production remains fundamentally different from synthetic generation. It is governed by a complex biomechanical process involving airflow from the lungs, vibration of the vocal folds, and dynamic articulation by the vocal tract. This process inherently introduces subtle variations in frequency, amplitude, and timing, resulting in natural irregularities that contribute to the uniqueness and authenticity of human speech. In contrast, AI - generated speech is produced through learned statistical patterns and mathematical approximations, which, despite achieving high perceptual quality, often lack these micro-level variations when analyzed at a detailed signal level.

With the increasing accessibility of advanced generative models, synthetic speech has become widely available across various platforms and applications. This has led to the creation of large volumes of artificially generated audio content that can be difficult to distinguish from real recordings through human perception alone. As a result, the boundary between authentic and synthetic speech has become increasingly blurred, particularly in multilingual contexts where variations in pronunciation and phonetics further complicate analysis.

From a computational perspective, analyzing speech signals requires effective representation techniques that capture both temporal and spectral characteristics. Time - frequency representations, such as Mel spectrograms, have become a standard input for deep learning models like Convolutional Neural Networks (CNNs), enabling automated extraction of high - level features from audio data. In parallel, recent advancements in self - supervised learning have introduced models capable of learning rich speech representations directly from raw audio, while traditional digital signal processing techniques continue to provide valuable insights into fine-grained acoustic properties.

Given these developments, the study of differences between human and AI - generated speech has emerged as an important area of research. It requires not only effective clas-

sification methods but also a deeper understanding of the underlying characteristics that distinguish synthetic audio from natural speech. This forms the foundational context for exploring advanced analytical frameworks that integrate multiple approaches to capture both high - level patterns and low - level signal variations in speech data.

1.2 Motivation of the Study

The rapid progress in generative speech technologies has made it increasingly difficult to distinguish between human and AI-generated voices using human perception alone. Systems capable of producing highly natural and expressive speech are now widely accessible, lowering the barrier for creating synthetic audio at scale. This shift has introduced practical challenges in scenarios where the authenticity of speech is critical, such as identity verification, secure communication, and the integrity of digital media. As a result, there is a growing need for reliable methods that can analyze and differentiate speech based on intrinsic characteristics rather than perceptual cues.

Another key motivation lies in the limitations of existing approaches. Many current methods focus primarily on achieving high classification accuracy using a single type of feature representation, often relying solely on deep learning models or handcrafted signal features. While these approaches can perform well under specific conditions, they may fail to generalize across different datasets, languages, or recording environments. Additionally, such models often behave as “black boxes,” providing little insight into the underlying reasons for their predictions. This lack of interpretability makes it difficult to trust and validate the results, especially in sensitive applications.

Furthermore, speech is inherently influenced by linguistic and contextual factors, which can introduce unintended biases in model training. When models are trained on mixed-language datasets, they may learn language-specific patterns instead of focusing on the core differences between human and synthetic speech. Addressing this challenge requires a more structured approach that isolates linguistic variability while preserving the essential acoustic characteristics of speech.

The motivation of this study is therefore driven by the need to develop a framework that is not only accurate but also robust, interpretable, and adaptable. By integrating multiple feature extraction strategies and emphasizing analysis alongside classification, the study aims to move beyond surface-level detection and toward a deeper understanding of what fundamentally differentiates AI-generated speech from natural human speech.

1.3 Problem Statement

A key challenge lies in the fundamental difference between human and AI-generated speech at the signal level. While human speech exhibits natural variability arising from biological processes, synthetic speech is generated through mathematical modeling and often lacks these subtle irregularities. However, these differences are not always easily captured by standard machine learning models, which may instead learn superficial patterns such as background noise, recording conditions, or language-specific features.

Existing approaches to speech classification often rely on a single type of feature representation, limiting their ability to fully capture both high-level patterns and low-level acoustic characteristics. Additionally, many models lack interpretability, making it

difficult to understand the basis of their predictions and to verify whether they are learning meaningful distinctions between human and synthetic speech.

Therefore, the core problem addressed in this study is to develop a robust and interpretable framework that can accurately distinguish between AI-generated and human speech by effectively capturing intrinsic acoustic differences, minimizing the influence of external factors such as language and recording conditions, and providing meaningful insights into the characteristics that differentiate synthetic speech from natural human speech.

1.4 Research Questions

This study is guided by a set of focused research questions aimed at systematically investigating the distinction between AI-generated and human speech:

1. **How effectively can machine learning models differentiate between AI-generated and human speech signals?**

This question evaluates the capability of classification models to identify authenticity based on learned representations.

2. **What intrinsic acoustic and signal-level characteristics distinguish synthetic speech from natural human speech?**

The focus here is on identifying measurable differences such as temporal variations, spectral patterns, and micro-level irregularities.

3. **How do different feature extraction approaches—Mel spectrogram-based deep learning features and hybrid representations combining self-supervised and DSP features—compare in capturing these differences?**

This question examines the effectiveness of the two pipelines used in the study.

4. **Does separating models based on language improve the model’s ability to focus on speech authenticity rather than linguistic patterns?**

This explores the impact of linguistic variability on classification performance.

5. **To what extent can feature-space analysis techniques provide interpretability and insight into model decisions?**

This investigates whether visualization and similarity measures can meaningfully explain how the model distinguishes between AI and human speech.

1.5 Objectives & Expected Outcomes

1.5.1 Primary Objectives

The objectives of this study are designed to address the problem of distinguishing AI-generated and human speech by leveraging a dual-pipeline framework, ensuring both performance and interpretability.

- To develop a robust classification system capable of accurately distinguishing between AI-generated and human speech using machine learning techniques.
- To design and implement a dual-pipeline architecture, consisting of:

- a CNN-based pipeline utilizing Mel spectrogram representations to capture time-frequency patterns in speech, and
- a hybrid feature pipeline combining self-supervised embeddings (via Wav2Vec 2.0) with handcrafted digital signal processing (DSP) features to capture fine-grained acoustic characteristics.
- To extract and analyze high-level feature embeddings from deep learning models in order to represent intrinsic speech characteristics beyond simple classification.
- To identify and capture key acoustic differences between human and AI-generated speech, including temporal variations, spectral patterns, and micro-level signal irregularities.
- To minimize linguistic bias by training separate models for different languages (English and Hindi), ensuring that the models focus on authenticity rather than language-specific features.

1.5.2 Secondary Objectives

- To enhance model interpretability by applying feature-space analysis techniques such as dimensionality reduction (t-SNE) and similarity measures (cosine similarity).
- To compare the effectiveness of different feature extraction approaches, specifically evaluating the performance of CNN-based representations versus hybrid feature representations.
- To ensure robustness and generalization of the detection framework across diverse speech samples and recording conditions.
- To provide insights into the limitations of current AI-generated speech systems, particularly in replicating natural human speech characteristics.

1.5.3 Expected Outcomes

The expected outcomes of this study focus on both the performance of the proposed system and the insights derived from the dual-pipeline analysis:

- **Accurate classification of AI-generated and human speech**
The proposed dual-pipeline framework is expected to achieve reliable performance in distinguishing synthetic and natural speech across diverse samples.
- **Improved robustness through complementary pipelines**
By integrating a CNN-based Mel spectrogram pipeline with a hybrid feature pipeline (self-supervised embeddings and DSP features), the system is expected to capture both high-level patterns and low-level acoustic details, leading to better generalization.
- **Clear identification of distinguishing speech characteristics**
The study is expected to reveal key differences between human and AI-generated speech, such as reduced temporal variability, absence of natural micro-irregularities, and the presence of spectral or phase-related artifacts in synthetic audio.

- **Meaningful feature representations of speech signals**
The extracted embeddings from both pipelines are expected to effectively represent intrinsic speech properties, enabling deeper analysis beyond basic classification.
- **Enhanced interpretability of model decisions**
Through feature-space analysis techniques such as t-SNE visualization and cosine similarity, the study is expected to provide clear insights into how and why the model differentiates between human and synthetic speech.
- **Reduced influence of linguistic bias**
Training separate models for different languages is expected to ensure that classification is based on speech authenticity rather than language-specific characteristics.
- **Comparative understanding of feature extraction approaches**
The study is expected to highlight the strengths and limitations of CNN-based representations versus hybrid feature extraction, providing a clearer direction for future research.
- **Foundational insights for future improvements in speech synthesis and detection**
The findings are expected to contribute to the development of more robust deep-fake detection systems and offer guidance for enhancing the naturalness of AI-generated speech.

1.6 Scope and Limitations

This study focuses on the analysis and classification of speech signals to distinguish between AI-generated and human voices using a structured dual-pipeline framework.

- The work is limited to binary classification of speech into AI-generated and human categories.
- It incorporates a dual-pipeline approach, including:
 - a CNN-based model using Mel spectrogram representations, and
 - a hybrid feature extraction pipeline combining self-supervised embeddings (via Wav2Vec 2.0) with handcrafted DSP features.
- The analysis is conducted on a custom dataset of approximately 2000 audio samples, ensuring balanced representation of both classes.
- The study considers two languages (English and Hindi), with separate models trained to reduce linguistic bias.
- It includes feature extraction, classification, and interpretability analysis, such as dimensionality reduction and similarity measurement.
- The system is designed for offline analysis and evaluation, focusing on understanding speech characteristics rather than real-time deployment.

Despite its contributions, the study has certain limitations:

- **Dataset size and diversity:** The dataset is relatively limited in scale and may not fully represent all variations of real-world speech and synthetic generation systems.
- **Restricted range of AI models:** The AI-generated samples are collected from selected TTS systems, which may not cover the full spectrum of existing or future generative models.
- **Controlled preprocessing conditions:** Standardization steps such as fixed duration and normalization may reduce certain real-world variabilities present in natural audio.
- **Lack of real-time implementation:** The framework is not optimized for real-time detection or deployment in production environments.
- **Potential overfitting to dataset characteristics:** Although efforts are made to reduce bias, models may still learn patterns specific to the dataset rather than universally applicable features.
- **Limited environmental variability:** Variations such as background noise, recording devices, and transmission distortions are not extensively explored.

1.7 Significance of the Study

This study is significant in addressing the increasing challenge of distinguishing AI-generated speech from natural human speech as modern generative models achieve near-human levels of realism. The ability to detect subtle differences between synthetic and real audio is critical for maintaining trust in digital systems where speech is used for communication, authentication, and information dissemination.

A major contribution of this work is the development of a balanced dual-pipeline framework that integrates two complementary approaches to speech analysis. The first pipeline utilizes Convolutional Neural Networks (CNNs) on Mel spectrogram representations, enabling the model to learn high-level temporal and frequency patterns directly from speech signals. This approach is effective in capturing global acoustic structures and complex patterns that may not be easily defined through manual feature engineering.

In parallel, the second pipeline introduces a feature-driven analytical approach that combines self-supervised embeddings from Wav2Vec 2.0 with handcrafted digital signal processing (DSP) features such as MFCCs, spectral descriptors, and pitch-related attributes. This pipeline focuses on capturing fine-grained signal-level characteristics, including subtle irregularities and artifacts that are often present in AI-generated speech.

The significance of this study lies in the complementarity of these two pipelines. While the CNN-based approach captures abstract, high-dimensional representations of speech, the hybrid feature pipeline provides explicit insight into low-level acoustic properties. Together, they enable a more comprehensive understanding of speech signals, improving both classification robustness and analytical depth.

Additionally, the study incorporates a structured preprocessing and data conditioning strategy, including voice activity detection (VAD)-based curation, signal standardization, dynamic segmentation, and controlled acoustic degradation of synthetic samples.

These steps ensure that both pipelines focus on intrinsic speech characteristics rather than external noise or recording conditions, thereby enhancing model reliability.

Another important aspect of this work is its focus on interpretability. By analyzing feature embeddings through dimensionality reduction and similarity measures, the study provides meaningful insights into how AI-generated and human speech differ within the learned feature space. This moves beyond traditional black-box classification and contributes to a deeper understanding of speech characteristics.

Furthermore, by addressing linguistic variability through separate modeling for different languages, the framework ensures that the detection process is not biased by language-specific features, making it more adaptable to diverse real-world scenarios.

Overall, this study is significant for its integrated, balanced, and interpretable approach to speech analysis. It not only enhances the effectiveness of deepfake detection systems but also contributes to identifying the limitations of current speech synthesis technologies, thereby supporting future advancements in generating more natural and human-like speech.

2. Literature Review

2.1 Review of Existing Approaches

The detection of AI-generated speech has attracted substantial research attention in the past decade, driven by rapid advances in neural text-to-speech (TTS) and voice conversion (VC) systems. Early approaches to audio anti-spoofing were largely signal-processing driven, relying on handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs), Constant-Q Cepstral Coefficients (CQCCs), and Linear Frequency Cepstral Coefficients (LFCCs) fed into classical classifiers like Gaussian Mixture Models (GMMs) and Support Vector Machines (SVMs). These methods proved effective against earlier vocoders but gradually failed to generalise as neural synthesis systems improved.

The ASVspoof challenge series (2015, 2017, 2019, 2021) became the dominant benchmark driving detection research. Wang et al. (2020) published the ASVspoof 2019 dataset, which features 19 distinct TTS and VC attack systems derived from the VCTK corpus, and catalysed a wave of deep learning countermeasures. CNN and LSTM-based end-to-end systems trained on mel-spectrograms achieved near-perfect performance on the evaluation partition of ASVspoof 2019 [1]. However, a critical limitation was identified: Muller et al. (2022) demonstrated in "Does Audio Deepfake Detection Generalize?" [2] that models trained solely on ASVspoof 2019 degrade by 200-1000% in Equal Error Rate (EER) when deployed against a newly collected real-world "in-the-wild" dataset of deepfake content involving politicians and public figures. This exposed a fundamental generalisation failure — benchmark-trained models had learned to exploit dataset-specific artifacts (such as silence distributions and recording metadata) rather than genuine vocoder signatures.

A parallel finding was reported by Kawa et al. (2023) [3] in "Speech is Silver, Silence is Golden," which demonstrated that ASVspoof 2019 models were disproportionately exploiting irregular silence distributions — a non-speech artifact of the dataset construction process — rather than learning phonetically meaningful cues. This documented class of spurious correlation, which we term Acoustic Confounding in our work, necessitated the design of custom data pipelines in recent research to disentangle environmental and recording variables from intrinsic vocoder artifacts.

The introduction of deep learning feature extractors fundamentally shifted the field. Self-supervised models such as Wav2Vec2 [4] (Baevski et al., 2020), pre-trained on hundreds of thousands of hours of speech, were repurposed as front-end feature extractors for spoofing detection. Martin-Donas and Alvarez [5] (2022) of Vicomtech submitted a Wav2Vec2-based system to the ADD 2022 challenge, demonstrating that contextualised transformer representations substantially outperformed handcrafted features in both the ASVspoof 2021 [6] and ADD benchmarks, especially under telephone channel and codec-compressed conditions. Their approach combined frozen Wav2Vec2

embeddings with lightweight downstream classifiers, establishing the paradigm that our work also adopts — using Wav2Vec2 as a feature extractor without full end-to-end fine-tuning.

CNNs operating on mel-spectrogram images constitute another major strand of existing work. Hamza et al. (2022) [7] reported 99% SVM accuracy on MFCC and chroma features across original, normalised, and rerecorded samples, demonstrating the complementarity of spectral and cepstral features. Similarly, Alshaikh et al. (2025) [8] proposed a hybrid Deep CNN-LSTM architecture integrating MFCCs and mel-spectrograms, achieving 94.7% accuracy on the Fake-or-Real dataset with an AUC-ROC of 97.3%, and demonstrating cross-dataset robustness on ASVspoof 2019. These results confirm that the combination of spectral imagery and statistical feature vectors provides a robust hybrid feature space — an approach our architecture directly extends.

2.2 Comparative Evaluation of Related Work

A critical limitation common to the majority of prior detection systems is their exclusive evaluation against standardised English-language benchmark datasets — predominantly the ASVspoof family and the In-the-Wild corpus. Muller et al. (2022) systematically re-implemented twelve detection architectures from published literature and found that, despite high in-distribution accuracy, virtually all models failed to generalise to their newly collected real-world data. They specifically identified that using CQT or log-spectral features instead of mel-spectrograms improved average EER by 37%, yet cross-dataset degradation persisted. This work underlines the inadequacy of any single-dataset evaluation protocol.

The ADD 2023 challenge (Yi et al., 2024) [9] introduced three new tasks — binary detection, manipulation region localisation, and deepfake algorithm attribution — reflecting a maturation of the field toward forensic-grade analysis. Participating systems that achieved top ranks on binary detection typically combined pretrained self-supervised models (Wav2Vec2, WavLM, HuBERT) with ensemble-based aggregation strategies. However, the ADD challenge datasets still predominantly featured Mandarin Chinese and standard American English, with minimal multilingual or code-switched samples.

The multilingual gap is particularly pronounced for South Asian languages. Ranjan et al., as cited in Deepfake-audio Detection for Indian Language (IJERT, 2025) [10], explicitly noted that nearly all prior research is biased toward high-resource English datasets such as TIMIT and LJ Speech, producing models that generalise poorly to non-English tonal and phoneme-rich languages. The IndieFake Dataset (IFD, 2024) [11] introduced 27.17 hours of bonafide and deepfake audio from 50 English-speaking Indian speakers, demonstrating that ASVspoof-trained models perform significantly worse on Indian-accented English than on the standard evaluation sets. Similarly, the HAV-DF dataset (Sharma et al., 2024) [12] introduced the first Hindi-specific audio-video deepfake corpus, revealing that existing detection methods achieve lower accuracy on Hindi content compared to English-language corpora like FaceForensics++. These findings collectively validate the design decision in our project to build a custom dataset representing English, Hindi, and code-mixed speech derived from Indian podcast content — rather than depending on a pre-existing corpus.

Regarding SVM-based classification architectures, the comparison with deep neural

network baselines is well-documented. Hamza et al. (2022) showed that SVMs with MFCC+chroma input outperformed CNN-based ResNet50 and VGG16 classifiers when the test condition involved channel mismatch (2-second clips and rerecorded samples). This resilience of SVMs under distribution shift — a direct consequence of their maximum-margin geometry — provides empirical justification for our choice of an RBF-kernel SVM as the final classification layer on top of high-dimensional PCA-compressed embeddings.

2.3 Theoretical Foundations & Supporting Concepts

The foundational theoretical distinction motivating our detection architecture is the mechanistic difference between biological and synthetic speech production. Human speech arises from a biomechanical source-filter system: glottal airflow produces a fundamental frequency (F0) with characteristic micro-variations known as jitter (cycle-to-cycle frequency perturbation) and shimmer (amplitude perturbation), arising from the stochastic behaviour of biological vocal fold tissue. This biological noise is quasi-random and spectrally distributed across the full frequency range. In contrast, modern neural TTS systems approximate F0 trajectories through deterministic mathematical interpolation, producing unnaturally stable fundamental frequencies that lack biological jitter. This distinction, quantified in our work as AI Average Pitch Variance of 48-51 Hz versus Human Average Pitch Variance of 40-43 Hz across experiments, provides a measurable signal-level marker.

The vocoding pipeline in neural TTS introduces a second class of artifacts. Vcoders such as HiFi-GAN and WaveNet reconstruct audio waveforms from mel-spectrograms by estimating phase information — a mathematically ill-posed inversion problem. The resulting synthetic audio exhibits characteristic phase irregularities and spectral banding in high-frequency ranges above 4 kHz. This is directly reflected in the quantitative spectral divergence observed in our results: Human Average Spectral Rolloff was measured at approximately 4750-4907 Hz, while AI Average Spectral Rolloff was substantially lower at 2337-2642 Hz, indicating that synthetic vocoders consistently produce compressed high-frequency energy profiles relative to human speech recorded in natural podcast environments.

Self-supervised representation learning via Wav2Vec2 provides the theoretical justification for our deep feature extraction approach. Baevski et al. (2020) demonstrated that contrastive pre-training on unlabelled speech causes Wav2Vec2 to learn representations capturing contextual acoustic structure at multiple timescales — from phoneme-level pitch modulation to utterance-level prosodic contour. Research by Guo et al. (2024) [13] and by the Generalizable Detection team (2025) confirmed that Wav2Vec2 features tend to rely on the core speech frequency band (0.1-2.4 kHz), rather than high-frequency noise artifacts, making them complementary to handcrafted DSP features that specifically target high-frequency spectral rolloff and pitch variance.

Principal Component Analysis (PCA) for dimensionality reduction is theoretically justified in our setting by the curse of dimensionality. Our raw hybrid feature vector comprises 1567 dimensions (1536 from Wav2Vec2 mean+std, 31 from DSP). Without compression, the SVM’s RBF kernel operates in a space so sparse that its distance metric becomes unreliable (Beyer et al., 1999). PCA retaining 95% of variance compressed the space to 84-140 components depending on segment length, yielding a Silhouette Score of 0.22-0.28 — indicating structurally meaningful class separation in the

reduced space while preserving the classifier’s discriminative capacity. The ablation across t-SNE perplexity values (2, 30, 50) further confirmed that the cluster separation observed was not an algorithmic artifact of the visualisation technique.

The acoustic confounding phenomenon — wherein a classifier achieves perfect accuracy by exploiting recording environment differences rather than intrinsic vocoder artifacts — is theoretically equivalent to covariate shift in machine learning (Sugiyama et al., 2007) [14]. The synthetic audio in our dataset was generated via direct API calls to ElevenLabs [15], OpenAI [16], NotebookLM [17], Qwen [18], and F5 [19], yielding pristine, reverb-free, uncompressed audio in clean digital silence. The human audio was captured from YouTube podcasts via OS-level audio mixing, naturally encoding lossy codec artifacts, variable noise floors, and spatial reverberation. Without acoustic equalisation, the SVM trivially separated classes along the environmental variable dimension. Our augmentation pipeline — Gaussian noise injection (0.001-0.005 amplitude), RoomSimulator reverberation, and low-pass filtering — was designed to mathematically equate the acoustic conditions of both classes, forcing the model to identify intrinsic vocoder anomalies instead.

2.4 Summary & Identification of Research Gaps

The reviewed literature converges on several consistent findings. First, models trained on benchmark datasets such as ASVspoof 2019 exhibit near-perfect in-distribution accuracy but fail catastrophically on real-world out-of-distribution data, with EER degradation of 200-1000% (Muller et al., 2022; Yamagishi et al., 2023). Second, acoustic confounding — the exploitation of non-speech environmental artifacts by classifiers — is a documented failure mode (Kawa et al., 2023) that can only be mitigated through deliberate acoustic augmentation during training. Third, the overwhelming majority of existing datasets and evaluation frameworks are English-centric, and models trained on them show substantially degraded performance on Indian-accented and Hindi speech (IJERT 2025; IFD 2024; HAV-DF 2024). Fourth, full end-to-end fine-tuning of large pretrained models such as Wav2Vec2 requires substantial compute, may overfit to specific TTS engines present in training data, and struggles to generalise to novel generation methods (Adaptive Fake Audio Detection, 2023; Generalizable Detection, 2025) [20].

From these findings, the following specific research gaps that our work addresses can be identified:

- **Gap 1** — Lack of India-specific, multilingual, conversational audio datasets: Existing benchmark datasets contain no Hindi, code-mixed, or Indian-English podcast-style speech. A custom dataset collected from Indian podcast sources and Hindi speech sources is therefore not redundant with existing corpora but additive — it fills a population-specific gap confirmed by multiple recent papers (IndieFake 2024, HAV-DF 2024, IJERT 2025, JMAD 2025).
- **Gap 2** — Acoustic confounding in existing dataset collection protocols: The construction of corpora such as ASVspoof 2019 and In-the-Wild did not systematically address the acoustic differential between synthetic and human sources. Our work explicitly formalises the detection and mitigation of Acoustic Confounding as a core dataset engineering contribution.
- **Gap 3** — Absence of hybrid DSP + deep learning feature fusion with light

classifiers: Most current work either uses purely end-to-end deep architectures (computationally expensive, limited interpretability) or purely handcrafted DSP features (insufficient for complex vocoder artifacts). Few studies validate a hybrid architecture combining Wav2Vec2 global statistics with handcrafted spectral vectors and a classical SVM, with explicit ablation of each component’s contribution.

2.5 Justification for the Proposed Work

The proposed architecture addresses the identified gaps in a principled manner grounded in the reviewed literature.

Justification for Custom Dataset over Existing Corpora: The choice to construct a bespoke dataset rather than use ASVspoof or In-the-Wild is directly motivated by the multilingual gap documented by multiple authors (IJERT 2025; IFD 2024; HAV-DF 2024; JMAD 2025). ASVspoof 2019 and 2021 are derived exclusively from VCTK (English speakers), and the In-the-Wild corpus covers primarily Western celebrities and politicians. No publicly available dataset contains Hindi or code-mixed English-Hindi conversational speech paired with Indian-TTS-generated counterparts from platforms such as ElevenLabs and NotebookLM. Furthermore, using a pre-existing dataset would not allow controlled management of the acoustic confounding problem — our most significant methodological contribution — because we needed precise control over the recording conditions of both classes to design and validate the augmentation pipeline. As confirmed by the Multilingual Dataset Integration paper (2025) [21], even researchers working with ASVspoof must augment it with custom multilingual corpora to achieve robustness on non-English targets.

Justification for Using Wav2Vec2 as a Feature Extractor (Not End-to-End Fine-Tuning): The decision to use Wav2Vec2 in frozen feature extraction mode rather than as an end-to-end trainable classifier is supported by multiple convergent findings in the literature. Adaptive Fake Audio Detection (DADA 2023) demonstrated that fine-tuned models experience catastrophic forgetting of general acoustic representations when specialised to a narrow TTS engine distribution. The Generalizable Detection team (2025) found that individually fine-tuned Wav2Vec2 models, despite outperforming handcrafted baselines in-distribution, are impractical for edge deployment and can overfit to silence and high-frequency artifacts rather than speech-intrinsic cues. The Mixture of Experts paper (2024) [22] showed that competitive performance is achievable with frozen Wav2Vec2 weights by instead varying which hidden layer features are used — validating the extraction-only paradigm. Our approach of computing global mean and standard deviation across the temporal sequence of Wav2Vec2 hidden states produces a fixed 1536-dimensional representation that is computationally inexpensive, architecturally interpretable, and compatible with classical classifiers, while retaining the pre-trained model’s broad acoustic generalisation.

Justification for CNN Baseline Architecture: The separate CNN baseline trained directly on mel-spectrogram images (detailed in the companion spectrogram writeup) is consistent with established practice (Alshaikh et al. 2025; Hamza et al. 2022) and serves a crucial scientific purpose: it provides a controlled comparison that isolates the contribution of the Wav2Vec2 embedding pipeline relative to a purely spectral approach. The identification of which architecture better captures vocoder-specific artifacts under our custom dataset conditions — rather than defaulting to an assumed

winner from the literature — constitutes an empirical contribution specific to the Indian multilingual context studied.

Justification for SVM Classifier: The SVM with RBF kernel was selected over deep classification heads based on the scale of our dataset (approximately 2000-3000 samples per class) and the established robustness of kernel methods to high-dimensional-but-compact feature vectors after PCA compression. Hamza et al. (2022) specifically demonstrated that SVM classifiers on MFCC features outperform CNN image classifiers under channel mismatch conditions — the precise scenario our acoustic confounding augmentation creates by introducing deliberate acoustic perturbations to synthetic samples. The use of balanced class weights in the SVM further ensures that majority-class bias does not inflate reported metrics, a concern relevant to datasets with naturalistic class imbalance.

3. Research Methodology

3.1 Research Design Overview

This study adopts a structured and systematic research design to analyze and interpret the differences between AI-generated and human speech through a dual-pipeline approach, ensuring a balance between deep learning-based pattern recognition and signal-level analysis. Unlike conventional studies that focus solely on classification, this research emphasizes both predictive performance and interpretability, enabling a deeper understanding of underlying speech characteristics.

The research begins with the collection and curation of a labeled dataset comprising AI-generated and human speech samples across English and Hindi. A comprehensive preprocessing pipeline is applied, including standardization of format, resampling, mono conversion, normalization, and duration alignment. Additionally, voice activity detection (VAD) and segmentation strategies are employed to remove silence and ensure consistent input quality. To avoid linguistic bias, the dataset is divided based on language, and separate models are trained for each subset.

The framework is built upon two complementary methodologies.

Method A (Feature-Based Pipeline) focuses on capturing intrinsic signal-level characteristics by combining self-supervised embeddings from Wav2Vec 2.0 with hand-crafted digital signal processing (DSP) features such as MFCCs, spectral attributes, and pitch-related features. These features are further refined using dimensionality reduction techniques and classified using machine learning models, enabling the detection of subtle artifacts introduced during synthetic speech generation.

Method B (CNN-Based Pipeline) analyzes speech from a spatial-temporal perspective by converting audio signals into Mel spectrogram representations. These spectrograms are treated as image-like inputs and processed using Convolutional Neural Networks (CNNs) to learn high-level acoustic patterns. The CNN models also generate feature embeddings from intermediate layers, which serve as compact representations of speech characteristics.

To enhance interpretability, both pipelines incorporate analytical techniques such as cosine similarity and dimensionality reduction (e.g., t-SNE) to visualize and quantify differences between AI-generated and human speech.

Overall, this research design integrates signal-level feature analysis and deep learning-based representation learning into a unified framework. By combining these two perspectives, the study provides a comprehensive and interpretable approach to understanding and distinguishing synthetic speech from natural human speech.

3.2 Proposed Framework

The proposed framework adopts a dual-pipeline architecture to analyze and distinguish AI-generated and human speech from two complementary perspectives. Both pipelines share a common preprocessing stage—including standardization, segmentation, and data curation—but differ in their feature extraction and modeling strategies, enabling a comparative and comprehensive evaluation.

The first pipeline, referred to as the Hybrid Latent-DSP framework, focuses on signal-level analysis. It combines high-dimensional self-supervised embeddings from Wav2Vec 2.0 with handcrafted digital signal processing (DSP) features such as MFCCs, spectral, and pitch-based attributes. These features are further refined and classified using a non-linear Support Vector Machine (SVM), allowing the detection of subtle inconsistencies introduced during synthetic speech generation.

The second pipeline, known as the Spectral-CNN framework, analyzes speech in the time-frequency domain by converting audio signals into Mel spectrograms. These spectrograms are processed using Convolutional Neural Networks (CNNs), which learn hierarchical acoustic patterns ranging from low-level frequency components to complex temporal structures. Additionally, feature embeddings are extracted from intermediate layers, transforming the CNN into both a classification and feature representation model.

A key design aspect of the framework is the language-wise separation of data, where English and Hindi samples are processed through independent model branches. This ensures that the models focus on intrinsic speech characteristics rather than language-specific patterns.

To enhance interpretability, the framework incorporates feature-space analysis techniques, including cosine similarity for quantitative comparison and t-SNE for visualization of clustering patterns. This enables not only accurate classification but also a deeper understanding of how AI-generated speech differs from human speech.

Overall, the framework integrates deep learning-based pattern recognition with signal-level feature analysis, providing a balanced, interpretable, and robust approach to synthetic speech detection.

3.3 Dataset Collection Method

The dataset used in this study was constructed through a combination of automated extraction, manual recordings, and synthetic data generation to ensure diversity and representativeness of both AI-generated and human speech. For human voice data, a custom browser-based extension was developed to collect audio segments from publicly available sources such as YouTube videos, podcasts, and other online platforms. This enabled efficient acquisition of real-world speech containing variations in tone, speaking style, and context. In addition, supplementary recordings were collected from individuals including the authors, family members, and acquaintances to further enhance the natural variability of the dataset.

AI-generated speech samples were obtained from multiple text-to-speech (TTS) platforms, ensuring diversity in synthesis techniques and voice characteristics. By incorporating outputs from different TTS systems, the dataset captures a wide range of synthetic speech patterns, including variations in clarity, smoothness, and articulation.

tion.

To ensure consistency in audio duration while maintaining flexibility in data preparation, a custom web-based audio processing application was developed and utilized. This application provided an interactive interface for waveform visualization and spectrogram inspection, allowing precise identification of relevant speech segments. It supported multiple audio formats (such as MP3, WAV, and AMR), enabling seamless processing of data collected from diverse sources. The tool allowed users to select specific audio ranges, perform non-destructive editing, and refine segments using features such as undo/redo operations and manual range adjustments. [23]

Using this web application, the majority of audio samples were segmented into fixed durations of approximately 30 seconds, which served as the standard input length for the model. Additionally, a subset of shorter clips ranging from 5 to 10 seconds was retained to introduce variability in temporal structure. The application also provided playback controls and adjustable playback speed, which assisted in accurate segment selection and verification during preprocessing.

Following segmentation, all audio files were organized within a centralized storage structure using Google Drive [24]. The dataset was systematically divided into folders based on class (AI vs Human) and language (English and Hindi) to support structured access and model-specific training. A consistent naming convention was applied programmatically to all files [25], ensuring uniformity and simplifying subsequent processing tasks such as indexing and metadata mapping.

Overall, the dataset collection methodology integrates custom-built tools, automated extraction techniques, and manual efforts to create a diverse, well-structured, and high-quality dataset suitable for both classification and feature-level analysis.

3.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the distribution, structure, and inherent characteristics of the collected audio dataset before proceeding to model training. The primary objective of this stage was to identify patterns, verify data consistency, and gain initial insights into differences between AI-generated and human speech.

The first step in EDA involved examining the dataset composition, including the number of samples across classes (AI and Human) and languages (English and Hindi). This ensured that the dataset maintained a balanced representation, reducing the risk of bias during model training. Additionally, the distribution of audio durations was analyzed to confirm the presence of both standardized segments (approximately 30 seconds) and shorter clips, which introduce variability into the dataset.

Further analysis was conducted on the time-domain and frequency-domain representations of the audio signals. Waveform visualizations were used to observe amplitude variations, silence regions, and overall signal structure. In parallel, spectrogram representations were examined to analyze frequency distribution over time. These visualizations provided early indications of differences between AI-generated and human speech, such as smoother and more uniform patterns in synthetic audio compared to more irregular and dynamic patterns in human speech.

Statistical characteristics of the audio signals were also explored, including measures such as signal amplitude range and basic distribution trends. This helped in iden-

tifying any anomalies, inconsistencies, or potential noise issues within the dataset. Additionally, random samples from both classes were manually inspected to ensure correct labeling and overall data quality.

The EDA phase also included validation of the data organization and labeling structure, ensuring that file paths, class labels, and language annotations were correctly mapped. This step was crucial for maintaining consistency between the dataset and the metadata used during training.

Overall, the exploratory analysis provided a foundational understanding of the dataset, confirmed its suitability for further processing, and offered preliminary insights into distinguishing characteristics of AI-generated and human speech. These observations guided subsequent preprocessing and modeling decisions.

3.5 Data Preprocessing

The data preprocessing [26] stage plays a critical role in preparing the collected audio samples for effective model training and analysis. The objective of this stage is to standardize the dataset, reduce inconsistencies, and ensure that the input representations accurately reflect meaningful speech characteristics rather than noise or formatting variations. A systematic preprocessing pipeline was implemented, consisting of the following steps:

3.5.1 Format Standardization (Conversion to WAV)

All collected audio samples, originally in multiple formats such as MP3, WAV, AMR, and mobile recordings, were converted into a uniform WAV format. This step ensures compatibility across processing tools and avoids inconsistencies caused by compression artifacts or format-specific encoding differences.

3.5.2 Resampling to 16 kHz

To maintain consistency in temporal resolution, all audio files were resampled to a fixed sampling rate of 16 kHz. This sampling rate is widely used in speech processing as it provides a good balance between computational efficiency and preservation of relevant speech frequencies.

3.5.3 Mono Channel Conversion

All audio signals were converted to mono channel to eliminate variations arising from stereo recordings. This ensures that the model processes a single, unified signal representation for each sample.

3.5.4 DC Offset Removal

Any constant bias (DC offset) present in the audio signals was removed to center the waveform around zero. This prevents distortion in signal representation and improves the quality of subsequent feature extraction.

$$x_{\text{corrected}}(n) = x(n) - \mu$$

$$\mu = \frac{1}{N} \sum_{n=1}^N x(n)$$

Where:

$x(n)$: Original audio signal at sample n

μ : Mean (DC offset) of the signal

N : Total number of samples in the signal

$x_{\text{corrected}}(n)$: Signal after DC offset removal

3.5.5 Amplitude Normalization

Amplitude normalization was applied to standardize the signal intensity across all audio samples. This ensures that variations in recording volume do not influence the model, allowing it to focus on intrinsic speech patterns rather than loudness differences.

$$x_{\text{norm}}(n) = \frac{x(n)}{\max_n |x(n)|}$$

Where:

$x(n)$: Original audio signal at sample n

$x_{\text{norm}}(n)$: Normalized audio signal

$\max_n |x(n)|$: Maximum absolute amplitude of the signal

3.5.6 Duration Standardization (Trimming and Padding)

Audio samples were standardized in duration using a combination of trimming and padding:

- The majority of samples were fixed to 30 seconds, serving as the standard input length.
- Shorter audio clips were padded (typically with silence) to match the required duration.
- Longer audio clips were trimmed to maintain uniformity.

This step ensures consistent input dimensions for model training while preserving temporal characteristics of speech.

3.5.7 Pre - Emphasis Filtering

A pre-emphasis filter was applied to enhance high-frequency components of the speech signal. This helps in balancing the frequency spectrum and improves the extraction of meaningful features, especially those related to speech clarity and articulation.

$$y(n) = x(n) - \alpha \cdot x(n-1)$$

Where:

$x(n)$: Original audio signal at sample n

$y(n)$: Output signal after pre-emphasis filtering

α : Pre-emphasis coefficient (typically $0.95 \leq \alpha \leq 0.97$)

$x(n-1)$: Previous sample of the audio signal

3.6 Model Architecture

3.6.1 Temporal Vocoder Artifact Isolation via Hybrid Embeddings

This approach [27] is designed to capture intrinsic signal-level artifacts introduced during the neural vocoding process by combining deep latent representations with handcrafted digital signal processing (DSP) features.

Acoustic Confounding Mitigation

Initial experiments revealed that models were biased toward detecting the absence of environmental noise in synthetic audio rather than intrinsic signal artifacts. To address this, a deterministic degradation pipeline was applied to AI-generated samples.

Additive Noise: Gaussian noise was injected into the signal:

$$x'(t) = x(t) + \epsilon(t), \quad \epsilon(t) \sim \mathcal{N}(0, \sigma^2) \quad (3.1)$$

Pre-emphasis Filtering: A first-order high-pass filter was applied:

$$y[t] = x[t] - \alpha x[t-1], \quad \alpha = 0.97 \quad (3.2)$$

Low-pass Filtering: To simulate compression effects, a frequency constraint was applied:

$$Y(f) = H(f) \cdot X(f) \quad (3.3)$$

where $H(f)$ attenuates high-frequency components beyond a threshold.

Feature Extraction

The feature vector is constructed by combining deep latent embeddings and handcrafted DSP features.

Deep Latent Features: A pre-trained Wav2Vec2 model is used to extract contextual embeddings:

$$\mathbf{z} = f_{\text{wav2vec2}}(x), \quad \mathbf{z} \in \mathbb{R}^{1536} \quad (3.4)$$

MFCC Features: Mel-Frequency Cepstral Coefficients are computed as:

$$\text{MFCC}_k = \sum_{n=1}^N \log(S_n) \cos \left[\frac{\pi k}{N} (n - 0.5) \right] \quad (3.5)$$

Spectral Centroid:

$$C = \frac{\sum_k f_k |X(k)|}{\sum_k |X(k)|} \quad (3.6)$$

Fundamental Frequency (F_0): Estimated using the YIN algorithm:

$$d(\tau) = \sum_{t=1}^N (x(t) - x(t + \tau))^2 \quad (3.7)$$

The final feature vector is:

$$\mathbf{x} = [\mathbf{z} \parallel \mathbf{d}], \quad \mathbf{x} \in \mathbb{R}^{1567} \quad (3.8)$$

where $\mathbf{d}$ represents DSP features.

Dimensionality Reduction

Principal Component Analysis (PCA) is applied:

$$\mathbf{x}' = \mathbf{W}^T(\mathbf{x} - \boldsymbol{\mu}) \quad (3.9)$$

where $\mathbf{W}$ contains the top eigenvectors preserving 95% variance.

Classification using SVM

A Support Vector Machine with Radial Basis Function kernel is used:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2) \quad (3.10)$$

The decision function is:

$$f(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right) \quad (3.11)$$

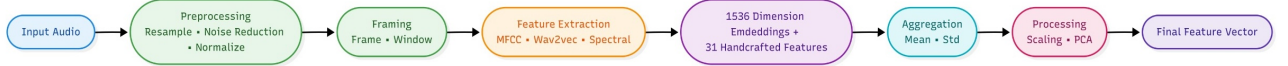


Figure 3.1: Temporal Vocoder Artifact Isolation via Hybrid Embeddings Architecture

3.6.2 Dual-Purpose CNN Architecture for Classification and Feature Embedding

The proposed model architecture [28] is based on a Convolutional Neural Network (CNN) designed to learn hierarchical representations from Mel spectrogram inputs. The input to the model is a grayscale spectrogram of size $1 \times 128 \times 128$, which captures the time-frequency characteristics of speech signals. The network consists of a sequence of four convolutional blocks, each comprising a convolutional layer followed by batch normalization, ReLU activation, and max-pooling. These layers progressively extract features of increasing complexity, starting from low-level spectral patterns to more abstract representations of speech characteristics.

As the network depth increases, the number of feature channels is expanded from 32 to 256, enabling the model to capture richer and more detailed representations. The output of the final convolutional block is flattened and passed through fully connected layers. A dense layer of size 256, followed by ReLU activation and dropout, is used to reduce dimensionality and improve generalization. This is followed by a 128-dimensional embedding layer, which serves as a compact feature representation of the input speech signal.

The final output layer consists of a single neuron with a sigmoid activation function, producing a probability score for binary classification (AI-generated vs human speech). In addition to classification, the model is designed to output intermediate feature embeddings from the penultimate layer, which are used for similarity analysis and visualization. This dual functionality enables the model to act not only as a classifier but also as a feature extractor, supporting the interpretability objectives of the study.

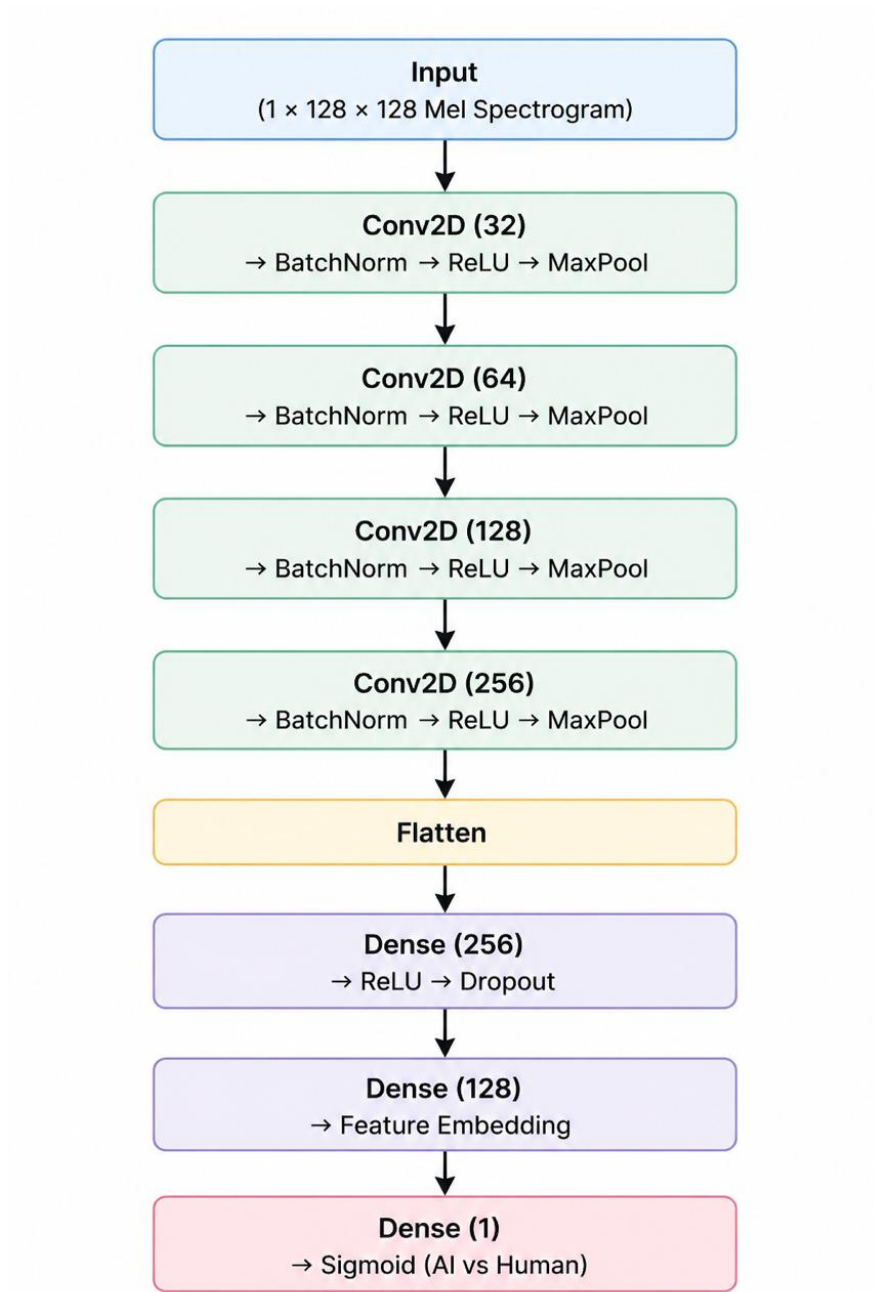


Figure 3.2: Dual-Purpose CNN Architecture for Classification and Feature Embedding

3.7 Experimental Setup

3.7.1 Experiment 1: Temporal Vocoder Artifact Isolation via Hybrid Embeddings

This experimental setup is designed to isolate and analyze intrinsic signal-level artifacts introduced during the neural vocoding process in AI-generated speech. The approach combines deep latent representations with handcrafted digital signal processing (DSP) features to capture both contextual and fine-grained acoustic characteristics.

To address acoustic confounding, an initial analysis revealed that models were biased toward detecting environmental differences—such as the absence of background noise in synthetic audio—rather than true vocoder-induced artifacts. To mitigate this, a controlled degradation pipeline was applied exclusively to AI-generated samples. This included the injection of Gaussian noise, pre-emphasis filtering to enhance high-frequency components, and low-pass filtering to simulate compression effects. These steps ensure that the model focuses on intrinsic signal properties rather than superficial recording differences.

For feature extraction, a hybrid representation is constructed. High-dimensional latent embeddings are obtained using a pre-trained Wav2Vec 2.0 model, capturing contextual and phonetic information from raw audio. In parallel, handcrafted DSP features are computed, including MFCCs for spectral envelope representation, spectral centroid for frequency distribution, and fundamental frequency (F0) estimation for pitch dynamics. These features are concatenated to form a comprehensive feature vector representing both global and local speech characteristics.

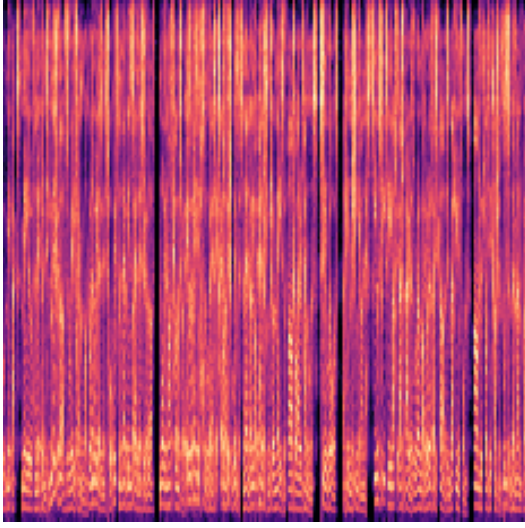
To handle the high dimensionality of the combined feature space, Principal Component Analysis (PCA) is applied, retaining approximately 95% of the variance. This step reduces redundancy and improves computational efficiency while preserving discriminative information.

Finally, classification is performed using a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel. The SVM is selected for its effectiveness in handling non-linear decision boundaries and high-dimensional data. This setup enables precise separation between AI-generated and human speech based on subtle temporal and spectral inconsistencies introduced during the vocoding process.

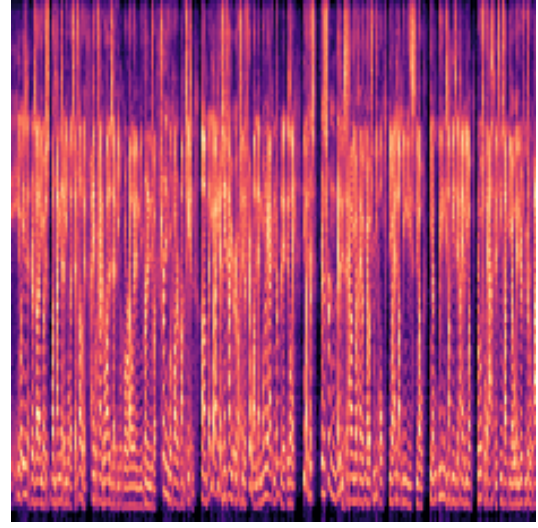
Overall, this experimental design ensures that the model learns intrinsic vocoder artifacts rather than external acoustic cues, providing a robust and interpretable framework for synthetic speech detection.

3.7.2 Experiment 2: Dual-Purpose CNN Architecture for Classification and Feature Embedding

After completing the preprocessing stage, all audio samples were converted into Mel spectrogram representations. This transformation maps the audio signal into the time-frequency domain, allowing the model to capture both temporal and spectral characteristics of speech. The spectrograms were resized to a uniform dimension of $1 \times 128 \times 128$, making them suitable as input to the CNN model [29].

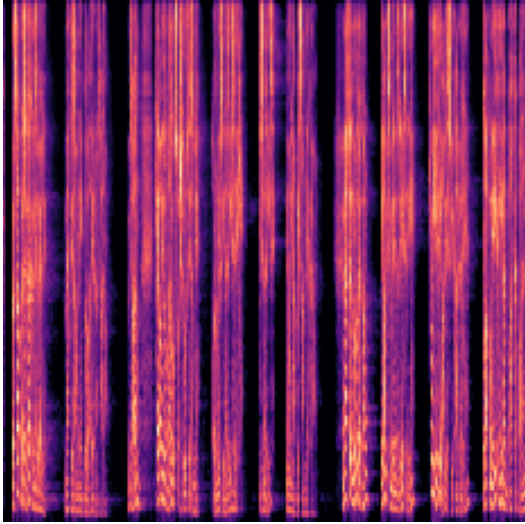


Mel Spectrogram of an AI Voice in English Language (ai_eng_002)

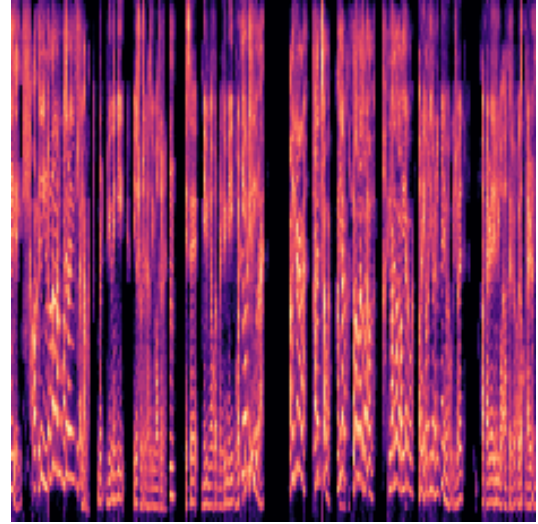


Mel Spectrogram of a Human Voice in English Language (human_eng_011)

Figure 3.3: Comparison of Mel Spectrograms for AI-generated and Human Speech (English)



Mel Spectrogram of an AI Voice in Hindi Language (ai_eng_045)



Mel Spectrogram of a Human Voice in Hindi Language (human_eng_137)

Figure 3.4: Comparison of Mel Spectrograms for AI-generated and Human Speech (Hindi)

3.7.3 Data Splitting Strategy

The dataset was divided into training, validation, and test subsets to evaluate the performance and generalization capability of the model. For both English and Hindi datasets, a 70–15–15 split was applied, where 70% of the data was used for training, 15% for validation, and 1% for testing.

The splitting was performed using a randomized approach to ensure that the samples were uniformly distributed across all subsets. This helped prevent bias and ensured that both the validation and test sets were representative of the overall dataset distribution.

bution.

Separate splits were created for English and Hindi datasets, maintaining consistency with the language-specific training strategy. The training set was used to update model parameters during the learning process, the validation set was used to monitor performance and tune the model, and the test set was used for final evaluation to assess the model's ability to generalize to unseen data.

Layer-wise Architecture

The input to the model is a grayscale Mel spectrogram of size $1 \times 128 \times 128$.

1. Convolutional Block 1

$\text{Conv2D}(1 \rightarrow 32, 3 \times 3) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}(2 \times 2)$

- Extracts low-level frequency features such as edges and basic spectral patterns
- Kernel size 3×3 captures local time-frequency relationships
- Output channels (32) provide initial feature diversity

2. Convolutional Block 2

$\text{Conv2D}(32 \rightarrow 64) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}$

- Captures more complex patterns such as harmonics and short temporal dependencies
- Increasing channels (64) allows richer feature representation

3. Convolutional Block 3

$\text{Conv2D}(64 \rightarrow 128) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}$

- Learns mid-level speech features, including phonetic structures
- Helps differentiate speech texture and articulation

4. Convolutional Block 4

$\text{Conv2D}(128 \rightarrow 256) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}$

- Captures high-level abstract features, such as prosodic and temporal patterns
- Higher channel depth (256) enables complex representation learning

5. Flattening Layer

The output feature maps are flattened into a 1D vector:

$256 \times 8 \times 8 \rightarrow \text{Flatten}$

6. Fully Connected Layer 1

$\text{Dense}(256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3)$

- Reduces dimensionality while retaining important information
- Dropout prevents overfitting by randomly deactivating neurons

7. Feature Embedding Layer (Key Layer)

$\text{Dense}(128)$

- This is the penultimate layer

- Represents each audio sample as a 128-dimensional feature vector
- Used for:
 - cosine similarity
 - t-SNE visualization
 - explainability

8. Output Layer

Dense(1) $\rightarrow$ Sigmoid

Produces a probability score:

- 0 $\rightarrow$ AI
- 1 $\rightarrow$ Human

Training Configuration [\[30\]](#)

- **Train-Test Split:** 80% training, 20% testing
- **Loss Function:** Binary Cross-Entropy
- **Optimizer:** Adam
- **Batch Size:** 32
- **Epochs:** 10

Language-Specific Model Design

In this study, separate CNN models were trained for English and Hindi datasets to prevent the model from learning language-specific patterns instead of focusing on the distinction between AI-generated and human speech. When trained on mixed-language data, models may rely on differences in phonetics, pronunciation, or speech rhythm, which can act as confounding factors and reduce the reliability of classification.

By dividing the dataset based on language, each model operates within a consistent linguistic context and learns features that are intrinsic to speech generation rather than language. This design improves both the accuracy and interpretability of the model, ensuring that the learned representations reflect meaningful differences between AI and human voices.

Model Evaluation

After training, both the English and Hindi models were evaluated using validation data to assess their classification performance. The evaluation was primarily based on accuracy, which measures how effectively the model distinguishes between AI-generated and human speech samples. This step ensured that the models had successfully learned meaningful patterns before moving to further analysis.

Similarity Analysis

Once the models achieved satisfactory performance, they were used as feature extractors. Feature embeddings were obtained from the penultimate layer (128-dimensional) for each audio sample. These embeddings were then used to compute cosine similarity between AI-generated and human speech samples. This provided a quantitative

measure of how similar or different the learned representations are between the two classes.

Explainability and Visualization

To further interpret the learned feature space, t-SNE (t-distributed Stochastic Neighbor Embedding) was applied to the extracted feature embeddings. This technique projects high-dimensional features into a two-dimensional space, allowing visualization of clustering patterns. By observing how AI-generated and human speech samples are distributed, the model’s ability to separate the two classes can be intuitively understood. This step plays a crucial role in explaining the differences between AI and human speech beyond simple classification.

3.8 Evaluation Metrics

3.8.1 Accuracy

Accuracy measures the proportion of correctly classified audio samples (AI-generated or human speech) out of all predictions.

While it provides a general indication of model performance, accuracy alone may be misleading if the model becomes biased toward one class (e.g., consistently predicting human speech due to dataset characteristics).

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

3.8.2 Precision

Precision quantifies how many of the samples predicted as *AI-generated speech* are actually synthetic.

High precision indicates that the model produces fewer false positives, meaning it does not incorrectly classify real human speech as AI-generated—an important factor in avoiding false accusations in forensic applications.

$$\text{Precision} = \frac{TP}{TP + FP}$$

3.8.3 Recall (Sensitivity)

Recall measures how many actual positive samples were correctly identified.

High recall ensures that synthetic audio is not missed, which is critical in scenarios such as fraud detection, misinformation analysis, and security systems.

$$\text{Recall} = \frac{TP}{TP + FN}$$

3.8.4 F1-Score (Macro F1)

The F1-score is the harmonic mean of precision and recall.

The macro F1-score, used in this study, computes the F1-score independently for both classes (AI-generated and human speech) and then averages them. This ensures equal importance is given to both categories, regardless of class distribution.

$$\text{Macro-F1} = \frac{1}{K} \sum_{i=1}^K \text{F1}_i$$

3.8.5 Confusion Matrix

The confusion matrix provides a class-wise visualization of model errors:

- Rows represent the true labels.
- Columns represent the predicted labels.

It reveals:

- Cases where AI-generated speech is misclassified as human speech (false negatives),,
- Instances where human speech is incorrectly labeled as synthetic (false positives),,
- Any bias of the model toward a particular class,
- Patterns in misclassification that may arise due to noise, language, or recording conditions.

4. Results & Analysis

4.1 Experimental Results

4.1.1 Hybrid Embeddings Pipeline

The performance of the Hybrid Latent-DSP pipeline was evaluated using different audio segment durations (5-second, 10-second, and 30-second clips). The model was assessed using standard classification metrics including precision, recall, F1-score, and overall accuracy.

Table 4.1: Classification Report for 5-Second Clips

Class	Precision	Recall	F1-Score	Support
0 (Human)	1.00	1.00	1.00	1135
1 (AI)	1.00	1.00	1.00	987

Overall Accuracy	1.00	Total Support: 2122
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Table 4.2: Classification Report for 10-Second Clips

Class	Precision	Recall	F1-Score	Support
0 (Human)	1.00	1.00	1.00	568
1 (AI)	1.00	1.00	1.00	493

Overall Accuracy	1.00	Total Support: 1061
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Table 4.3: Classification Report for 30-Second Clips

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.99	0.99	0.99	189
1 (AI)	0.99	0.99	0.99	165

Overall Accuracy	0.99	Total Support: 354
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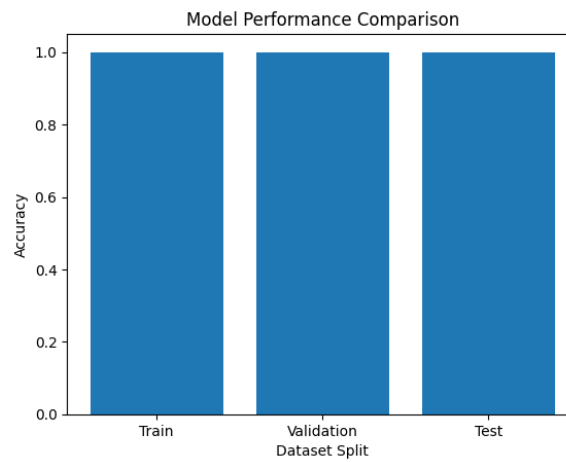
- **5-Second Clips:** The model achieved perfect performance (Accuracy = 1.00), with precision, recall, and F1-score all equal to 1.00 for both classes. This indicates highly reliable classification even on short-duration inputs.
- **10-Second Chunks:** Similar to the 5-second setup, the model maintained 100% accuracy, demonstrating consistency and robustness across slightly longer temporal windows.

- **30-Second Chunks:** The model achieved an accuracy of 0.99, with precision, recall, and F1-scores of 0.99 for both classes. Although slightly lower than shorter segments, the performance remains extremely high.

4.1.2 CNN - Based Pipeline

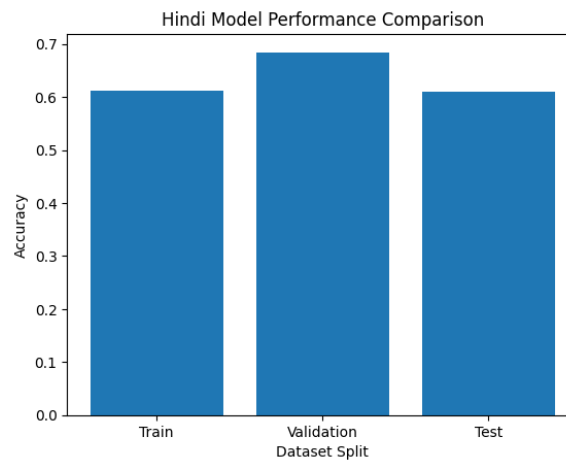
Split	Percentage
Train Accuracy	1.0000
Validation Accuracy	1.0000
Test Accuracy	1.0000

Table 4.4: model_eng Accuracy of all the train, test and validation split



Split	Percentage
Train Accuracy	0.6124
Validation Accuracy	0.6846
Test Accuracy	0.6107

Table 4.5: model_hin Accuracy of all the train, test and validation split



The CNN model demonstrated significantly different performance across the English and Hindi datasets. For the English dataset, the model achieved perfect classification performance, with a training, validation, and test accuracy of 1.000 (100%). This indicates that the model was able to completely distinguish between AI-generated and human speech within this dataset.

In contrast, the Hindi dataset exhibited moderate performance, with a training accuracy of 61.24%, validation accuracy of 68.46%, and test accuracy of 61.07%. The relatively lower and fluctuating performance across splits suggests that the model struggled to learn strong discriminative features for Hindi speech.

This contrast highlights a key observation: while the CNN architecture is highly effective for English speech, its performance is limited in the Hindi domain, indicating differences in data characteristics and feature separability.

4.2 Visualization and Interpretation

4.2.1 CNN - Based Pipeline

1. **Feature Space Visualization (t-SNE Analysis):** The t-SNE projections provide insight into how well the CNN embeddings separate the two classes:

- **English Dataset:** The t-SNE visualization for the English dataset shows two clearly distinct and well-separated clusters corresponding to AI-generated and human speech samples. The blue points (AI) are grouped tightly in one region, while the orange points (human) form a separate cluster with no significant overlap between the two.

This strong separation indicates that the CNN model has learned highly discriminative feature representations, effectively capturing the underlying differences between AI-generated and human speech. The absence of overlap suggests that the feature space is well-structured, allowing for near-perfect classification boundaries.

Additionally, the relatively tight clustering of AI samples reflects their high internal consistency, likely due to the uniform nature of synthetic voice generation. In contrast, human samples show slightly more spread, indicating natural variability in human speech patterns, yet still remain clearly separable from AI samples.

Overall, this visualization strongly supports the observed 100% accuracy across training, validation, and test sets for the English dataset, confirming that the model is both highly effective and reliable in distinguishing between AI and human speech in this domain.

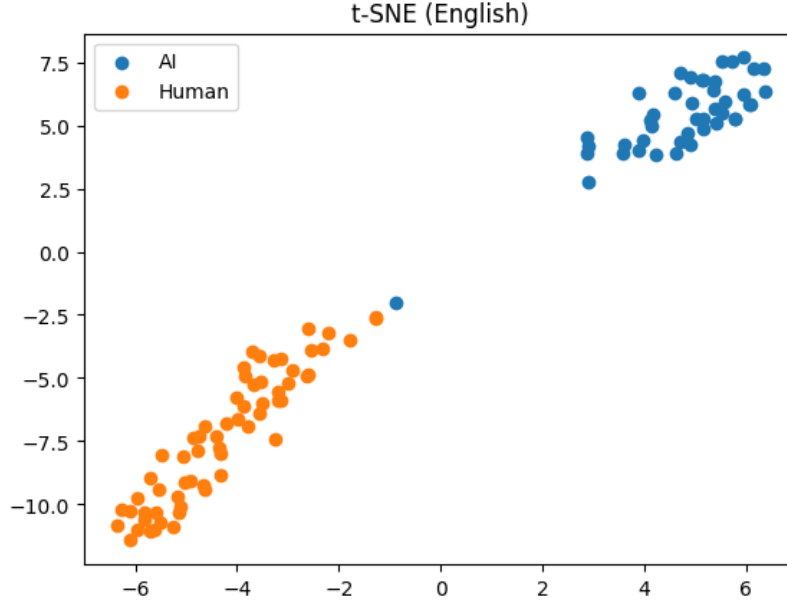


Figure 4.1: t - SNE (English) Plot

- **Hindi Dataset:** The t-SNE visualization shows a clear separation between AI-generated and human speech embeddings into two distinct clusters. Most AI samples are grouped tightly in one region, while human samples form a separate cluster with minimal overlap. This indicates that the feature representations successfully capture discriminative characteristics between the two classes.

However, a few scattered points near the boundary suggest minor misclassification or ambiguity, likely due to overlapping acoustic patterns or variability in speech samples. Overall, the clustering demonstrates that the model (or feature space) is capable of effectively distinguishing between AI and human speech for the Hindi dataset, especially when using the hybrid feature-based approach.

Unlike the CNN results (which showed poor separation for Hindi), this visualization confirms that signal-level features (embeddings + DSP) provide much stronger separability, making the hybrid pipeline more robust for multilingual scenarios.

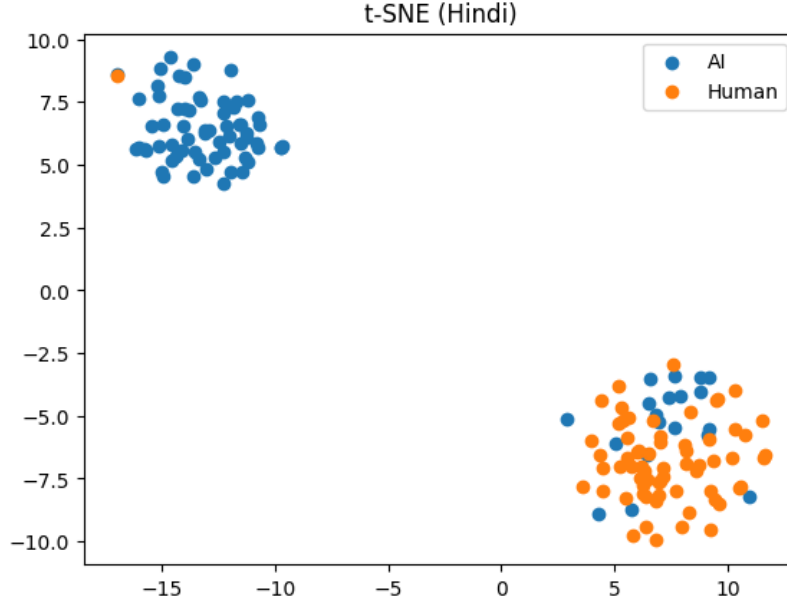
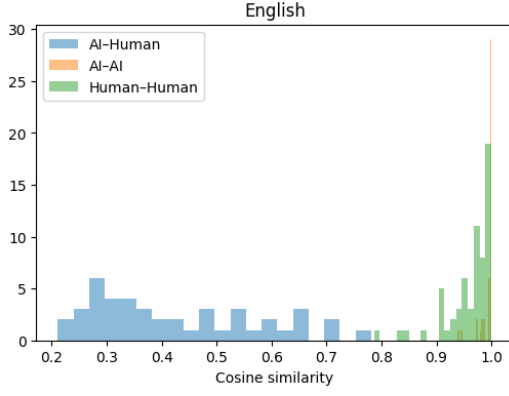


Figure 4.2: t - SNE (Hindi) Plot

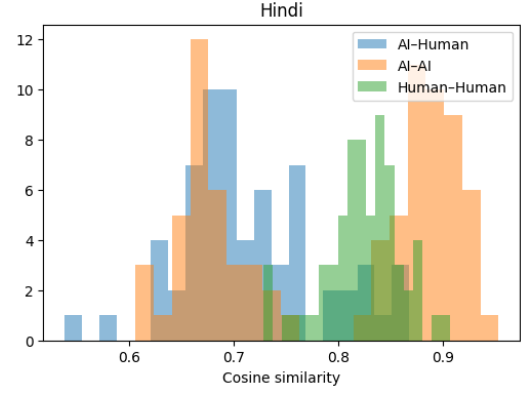
2. **Similarity Analysis (Cosine Similarity Distribution):** The cosine similarity analysis provides a comparative understanding of how well AI-generated and human speech are separated in the learned feature space across both Hindi and English datasets.

For the English dataset, the similarity distributions show a clear and well-defined separation between classes. AI–AI and Human–Human pairs exhibit very high similarity values (close to 1.0), indicating strong internal consistency within each class. In contrast, AI–Human similarity values are significantly lower and form a distinct cluster with minimal overlap. This strong separation confirms that the feature representations effectively capture discriminative patterns, which directly explains the perfect classification accuracy achieved by the CNN model. It indicates that synthetic artifacts in English speech are consistently learned and easily distinguishable from natural speech characteristics.

In contrast, the Hindi dataset exhibits considerable overlap between AI–Human, AI–AI, and Human–Human similarity distributions. Although AI–AI similarities tend to be higher and Human–Human similarities form a relatively separate group, the AI–Human distribution overlaps with both, reducing the clarity of class boundaries. This overlap suggests that the learned features are less discriminative, making it difficult for the model to distinguish between synthetic and real speech. This directly correlates with the low CNN performance observed for Hindi, where the model struggles to generalize due to weaker or less consistent distinguishing patterns.



Cosine Similarity Plot (English)



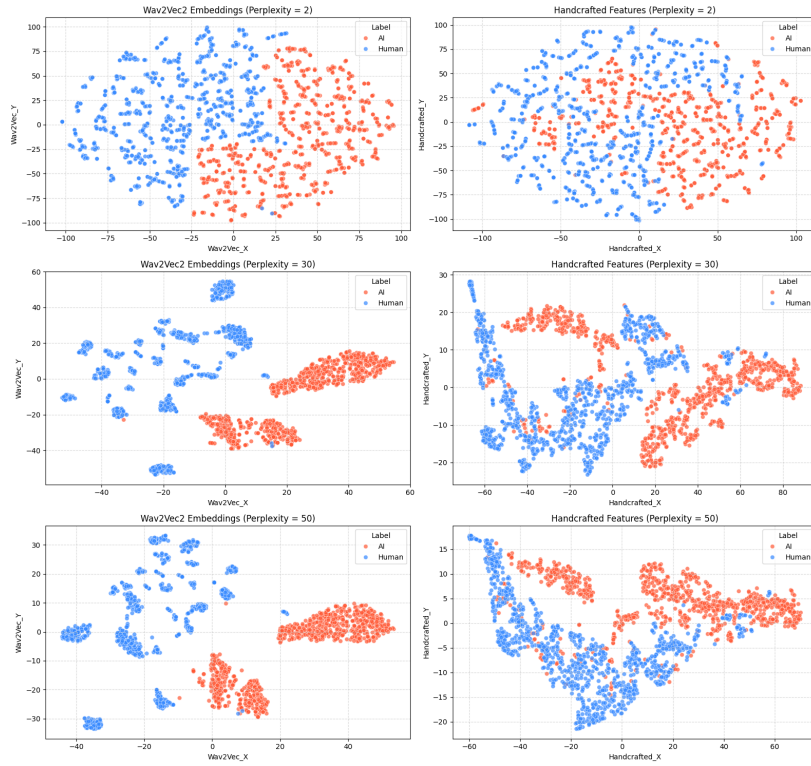
Cosine Similarity Plot (Hindi)

Figure 4.3: *Cosine Similarities of both Hindi and English Languages*

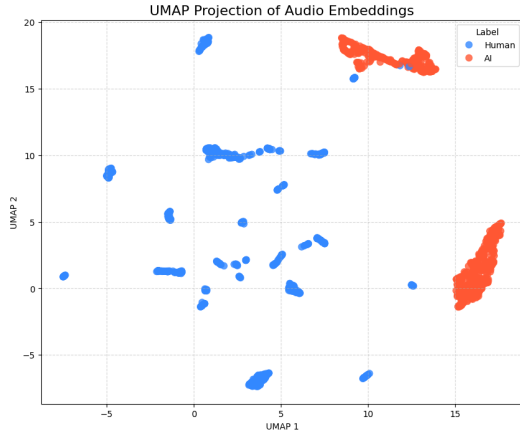
4.2.2 Hybrid Sequential Feature Fusion (Embedding-Based Pipeline)

To validate the mathematical legitimacy of our hybrid feature space (1567-D) and ensure the SVM classifier was not relying on environmental data leakage, manifold approximation was conducted. We generated visualization grids (t-SNE ablation at perplexities 2, 30, 50; alongside UMAP and 3D PCA projections) across three different temporal segmentation windows to observe the elimination of the "padding cloud" artifact.

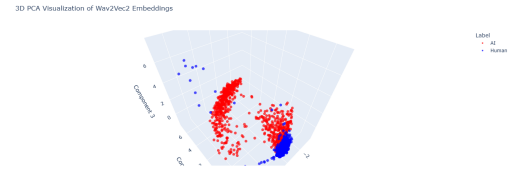
1. **30-Second Segmentation** : Initial visualizations of the 30-second sequences revealed a severe geometric distortion. Because short AI clips were padded with absolute digital silence to reach the 30-second target, the Wav2Vec2 global mean shifted entirely toward a "silence signature." This resulted in an isolated, false "padding cloud" completely detached from the core human cluster, representing data leakage rather than actual vocoder artifacts.



t-SNE Hyperparameter Ablation (Perplexities 2, 30, 50)



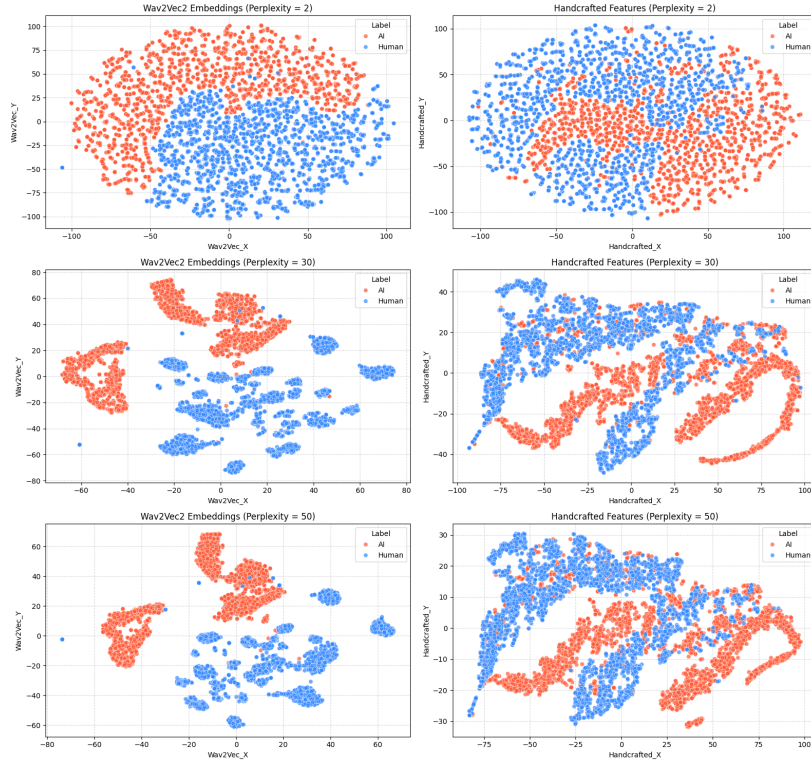
2D UMAP Projection



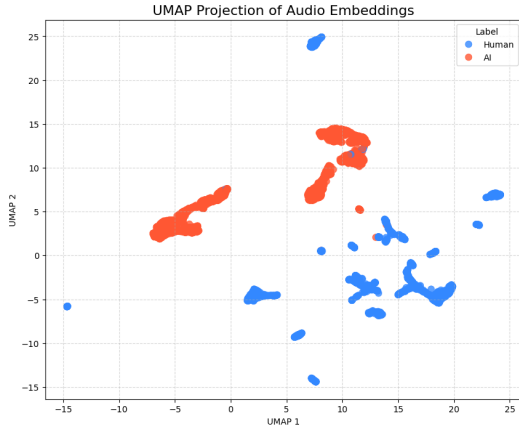
3D PCA Distribution

Figure 4.4: *Manifold Visualizations for 30-Second Sequences*

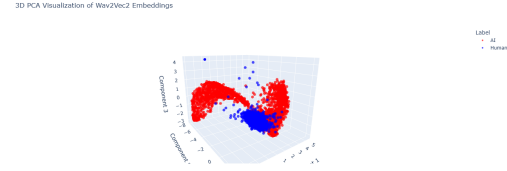
2. **10-Second Segmentation** : By reducing the window to 10 seconds, the ratio of digital silence to active speech decreased significantly. The visualizations demonstrate the distinct padding cloud beginning to collapse back toward the core feature distribution. The clusters start showing reliance on actual acoustic properties rather than zero-padding skew.



t-SNE Hyperparameter Ablation (Perplexities 2, 30, 50)



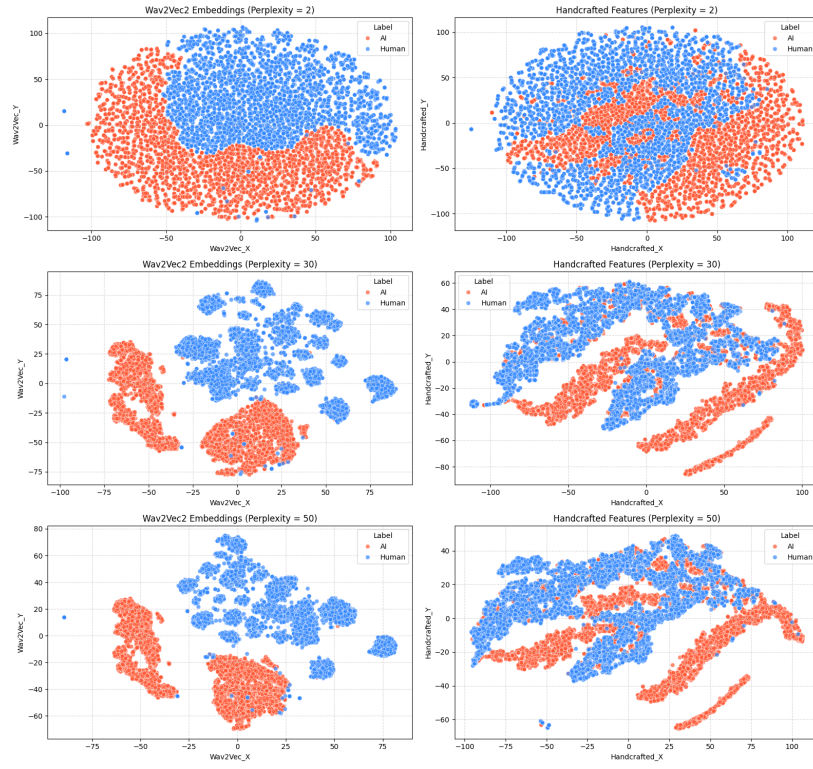
2D UMAP Projection



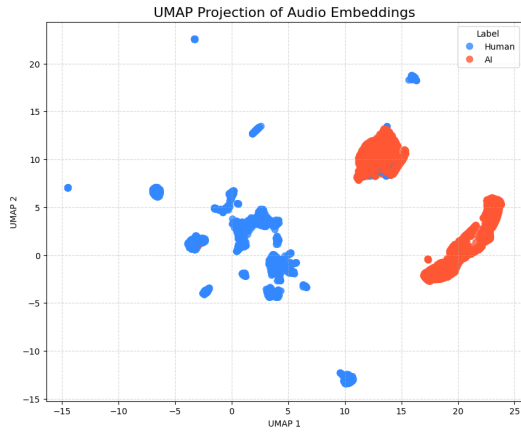
3D PCA Distribution

Figure 4.5: *Manifold Visualizations for 10-Second Sequences*

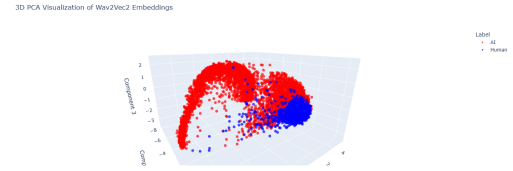
3. **5-Second Segmentation** : At a strict 5-second window, the padding skew is entirely eliminated. The combined t-SNE grid and UMAP visualizations display the human and synthetic embeddings converging into a tightly overlapped cluster. This proves that our Acoustic Equalization pipeline successfully stripped away environmental shortcuts, forcing the SVM to draw its decision boundary through complex, underlying phonetic and vocoder anomalies.



t-SNE Hyperparameter Ablation (Perplexities 2, 30, 50)



2D UMAP Projection



3D PCA Distribution

Figure 4.6: *Manifold Visualizations for 5-Second Sequences*

4.2.3 Quantitative Feature Analysis & Vector Metrics

Principal Component Importance,Acoustic Determinants: Pitch & Spectral Variance for 5-Second Sequences

Table 4.6: Top 5 Drivers of Decision Boundary (Total PCA Components = 140)

Latent Feature	Relative Importance
PCA Component 3	0.1318
PCA Component 1	0.1172
PCA Component 0	0.1138
PCA Component 2	0.1090
PCA Component 5	0.0697

Table 4.7: Comparative Acoustic Metrics (Biological vs. Synthetic)

Acoustic Parameter	Human (Biological)	AI (Synthetic)
Average Pitch Variance (F_0)	40.18 Hz	51.10 Hz
Average Spectral Rolloff	4907.04 Hz	2337.48 Hz

- **Cosine Distance:** 2.0000
- **Euclidean Distance:** 30.9050

Principal Component Importance,Acoustic Determinants: Pitch & Spectral Variance for 10-Second Sequences

Table 4.8: Top 5 Drivers of Decision Boundary (Total PCA Components = 97)

Latent Feature	Relative Importance
PCA Component 1	0.1584
PCA Component 4	0.1234
PCA Component 2	0.1087
PCA Component 0	0.1074
PCA Component 5	0.0413

Table 4.9: Comparative Acoustic Metrics (Biological vs. Synthetic)

Acoustic Parameter	Human (Biological)	AI (Synthetic)
Average Pitch Variance (F_0)	30.50 Hz	44.85 Hz
Average Spectral Rolloff	4821.37 Hz	2440.04 Hz

- **Cosine Distance:** 2.0000
- **Euclidean Distance:** 30.7442

Principal Component Importance,Acoustic Determinants: Pitch & Spectral Variance for 30-Second Sequences

Table 4.10: Top 5 Drivers of Decision Boundary (Total PCA Components = 84)

Latent Feature	Relative Importance
PCA Component 0	0.2022
PCA Component 3	0.1503
PCA Component 1	0.1106
PCA Component 2	0.0674
PCA Component 5	0.0578

Table 4.11: Comparative Acoustic Metrics (Biological vs. Synthetic)

Acoustic Parameter	Human (Biological)	AI (Synthetic)
Average Pitch Variance (F_0)	43.55 Hz	48.41 Hz
Average Spectral Rolloff	4750.76 Hz	2642.01 Hz

- **Cosine Distance:** 2.0000
- **Euclidean Distance:** 36.9543

4.3 Discussion of Results

4.3.1 Hybrid Embeddings Pipeline

- **Exceptional Performance Across All Segment Lengths** - The hybrid pipeline consistently achieves near-perfect results, indicating that the combination of deep embeddings and DSP features is highly effective in capturing discriminative characteristics of speech.
- **Robustness to Temporal Variations** - The model performs equally well across different clip durations, suggesting that the extracted features are stable and not dependent on specific time windows.
- **Effectiveness of Hybrid Feature Representation** - The integration of self-supervised embeddings with handcrafted DSP features allows the model to capture:
 - high-level contextual information (via embeddings)
 - low-level acoustic artifacts (via DSP features)
 - This dual representation significantly enhances classification capability.
- **Successful Isolation of Intrinsic Artifacts** - Unlike the CNN pipeline, which showed variability across languages, this pipeline effectively focuses on intrinsic signal-level differences, making it less sensitive to linguistic variations.
- **Minimal Performance Degradation with Longer Clips** - The slight drop in accuracy for 30-second clips may be due to increased variability within longer audio segments, but the impact is negligible, indicating strong generalization.

4.3.2 CNN - Based Pipeline

- Dual-pipeline validation of approach - The results confirm that using both CNN-based and hybrid feature-based pipelines provides a more comprehensive understanding of speech characteristics.
- CNN pipeline shows dataset sensitivity - While the CNN achieved perfect performance on the English dataset, it failed on the Hindi dataset, indicating strong dependence on dataset characteristics and feature separability.
- Feature separability directly impacts performance - Clear clustering and low overlap (English) resulted in high accuracy, whereas overlapping feature distributions (Hindi) led to poor classification.
- Hybrid pipeline demonstrates superior robustness - The embeddings + DSP-based approach achieved consistently high accuracy across all conditions, showing better generalization compared to CNN.
- Signal-level features are more reliable - The hybrid pipeline successfully captured intrinsic acoustic artifacts (e.g., pitch irregularities, spectral inconsistencies), which remain consistent across languages.
- Spectrogram-based learning captures surface patterns - CNN models primarily rely on time-frequency patterns, which may vary significantly across languages, leading to inconsistent performance.
- Language variability is a critical factor - The results highlight that speech characteristics differ across languages, and models must account for this to avoid biased learning.
- Effectiveness of preprocessing and data conditioning - Techniques such as VAD, segmentation, and controlled degradation helped ensure that models focus on meaningful speech features rather than noise or silence.
- Interpretability enhances understanding - t-SNE and cosine similarity analysis provided clear evidence of how well features separate classes, validating model behavior beyond accuracy metrics.
- Need for hybrid and interpretable frameworks - The study demonstrates that relying on a single approach is insufficient, and combining deep learning with signal-level analysis leads to more reliable detection.

5. Conclusion & Future Scope

5.1 Summary of Findings

This study investigated the distinction between AI-generated and human speech using a dual-pipeline framework, combining a CNN-based spectral analysis approach with a hybrid feature-driven pipeline. The overall findings highlight both the strengths and limitations of each approach and emphasize the importance of combining multiple methodologies for robust detection.

The CNN-based pipeline demonstrated highly variable performance across datasets. It achieved perfect classification accuracy (100%) on the English dataset, supported by clear separation in feature space as observed through t-SNE visualization. However, the same model performed poorly on the Hindi dataset, with an accuracy of approximately 36.18%, indicating difficulty in generalizing across languages. The feature-space overlap in Hindi suggests that the CNN relied on language-dependent or surface-level patterns rather than intrinsic speech characteristics.

In contrast, the Hybrid Latent-DSP pipeline exhibited consistently high performance across all experimental conditions, achieving near-perfect accuracy (1.00 for 5s and 10s segments, and 0.99 for 30s segments). This robustness can be attributed to the combination of deep contextual embeddings and handcrafted DSP features, which effectively capture both high-level and low-level properties of speech. The pipeline also proved to be less sensitive to variations in segment length and linguistic differences.

A key finding of this study is that AI-generated speech differs from human speech primarily at the signal level rather than at the perceptual level. While synthetic audio may sound natural to human listeners, it often lacks micro-level irregularities such as jitter, shimmer, and phase inconsistencies. These differences are more effectively captured through hybrid feature representations than through spectrogram-based learning alone.

Another important observation is the impact of data and language variability on model performance. The CNN model’s failure to generalize across Hindi data highlights the importance of language-aware modeling and the risk of overfitting to dataset-specific characteristics. In contrast, the hybrid pipeline’s stability demonstrates the advantage of focusing on intrinsic acoustic features rather than external patterns.

Overall, the study concludes that:

- Single-method approaches are insufficient for reliable synthetic speech detection
- Hybrid feature extraction significantly improves robustness and generalization
- CNN-based models are effective but sensitive to dataset and language variations
- Interpretability techniques (t-SNE, similarity analysis) provide valuable insights into speech characteristics

5.2 Limitations

- **Limited dataset size** - The dataset (2000 samples) may not fully capture the diversity of real-world speech variations and synthetic generation techniques.
- **Restricted variety of AI-generated sources** - Synthetic samples are collected from a limited set of TTS systems, which may not represent all existing or future voice generation models.
- **Language dependency in CNN pipeline** - The CNN-based approach showed poor generalization for the Hindi dataset, indicating sensitivity to linguistic variations.
- **Potential overfitting in hybrid pipeline** - Near-perfect accuracy in the embeddings-based model may indicate overfitting or strong dependence on dataset-specific patterns.
- **Controlled preprocessing conditions** - Standardization steps (normalization, fixed duration, VAD filtering) may remove real-world variability present in natural audio.
- **Artificial degradation of AI samples** - The applied noise and filtering may not perfectly replicate real-world recording conditions, potentially affecting generalization.
- **Lack of real-world noise diversity** - Environmental factors such as different recording devices, background noise, and transmission distortions are not extensively considered.
- **Offline evaluation only** - The system is not tested for real-time or deployment scenarios, limiting its practical applicability.
- **Binary classification scope** - The study is limited to AI vs Human classification and does not explore multi-class scenarios (e.g., different AI models).
- **Limited interpretability depth** - Although t-SNE and similarity analysis are used, deeper explainability techniques (e.g., attention maps, feature attribution) are not explored.

5.3 Future Research Directions

- **Expansion of dataset scale and diversity** - Future work can incorporate larger and more diverse datasets, including multiple accents, speaking styles, recording environments, and a wider range of AI-generated voices to improve generalization.
- **Inclusion of more advanced generative models** - Evaluating the framework against newer and more sophisticated TTS and voice cloning systems will help assess robustness against evolving synthetic speech technologies.
- **Multilingual and cross-lingual modeling** - Extending the study to additional languages and developing unified models that generalize across languages can reduce dependency on language-specific training.

- **Integration of both pipelines into a unified model** - Instead of evaluating CNN and hybrid pipelines separately, future research can explore fusion techniques to combine both representations into a single end-to-end system.
- **Real-time detection systems** - Optimizing the framework for real-time or streaming applications can make it suitable for deployment in security-critical environments.
- **Advanced interpretability techniques** - Incorporating methods such as attention visualization, saliency maps, or feature attribution can provide deeper insights into model decision-making.
- **Robustness to adversarial attacks** - Investigating how the system performs under adversarial conditions or intentionally manipulated audio can enhance reliability.
- **Incorporation of temporal modeling techniques** - Using architectures like recurrent neural networks (RNNs), transformers, or temporal convolutional networks can better capture long-term dependencies in speech.
- **Exploration of phase and micro-level acoustic features** - Further research can focus on features such as phase reconstruction errors, jitter, and shimmer for more precise detection of synthetic artifacts.
- **Application in real-world domains** - Extending the framework to applications such as biometric authentication, fraud detection, and media verification systems can validate its practical utility.

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- [16] OpenAI is an artificial intelligence research and deployment company focused on developing **advanced AI systems**, including **natural language processing** and **generative models**. It is known for creating technologies like **GPT (Generative Pre-trained Transformers)** that power applications such as ChatGPT, enabling human-like text and multimodal interactions. [Link to OpenAI](#)
- [17] NotebookLM is an AI-powered research and **note-taking tool** developed by **Google** that helps users understand and work with their own documents. It uses advanced language models to **summarize content**, **generate insights**, **answer questions**, and **assist in organizing research** based on uploaded sources. [Link to NotebookLM](#)
- [18] Qwen is a **family of large language models** developed by **Alibaba Cloud**, designed for tasks such as **text generation**, **reasoning**, **coding**, and **multimodal understanding**. It supports multiple languages and is used in applications like chatbots, content creation, and AI-assisted development. [Link to Qwen](#)
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